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AI3403: Multi-Agent Systems (MAS) - Assignment #3

Rules for Submitting Assignment

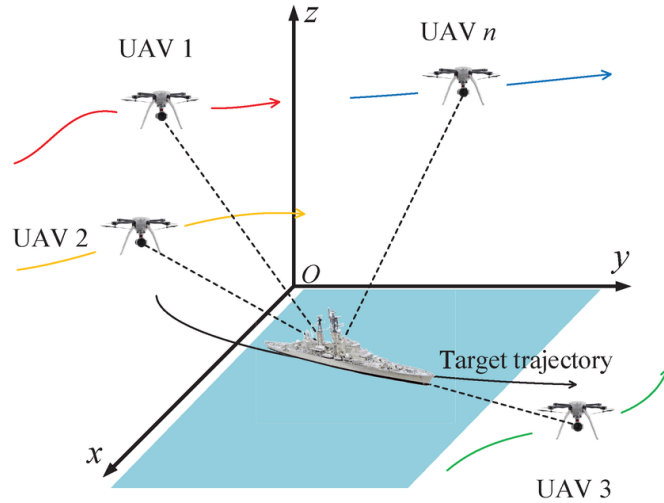
- The assignment is worth 100 points. You must present the entire work leading to their solution and also attach the associated code.
 - The assignment will be due by **18:00 on Friday, September 18, 2026** in the Google classroom. The deadline is strict and no extensions shall be awarded unless otherwise for unavoidable circumstances.
 - Usage of AI tools like ChatGPT/Claude/Gemini is permitted and must be limited only to understand the concepts taught in the lecture. If you use them in the assignment, you should explicitly declare where and to what extent you have used it. Educate yourself enough to know and spot when the AI tool that you are using is giving you some nonsense!
 - You should submit your assignment as **one** pdf document and the file should be named using roll number. For instance, if your roll number is **ai25btech11111**, then the pdf should be named as **ai25btech11111Assignment1.pdf**. Please also write your name in the first page of your assignment.
 - Students are strongly encouraged to submit their work in a neatly typed $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ document with all supporting code attached to the same document. An example overleaf latex template can be found in this link: <https://www.overleaf.com/latex/templates/assignment-homework-solutions-template/nfjybpqbdrn>. If typing is not possible, hand-written submissions will be considered as long as they are legibly written and scanned properly with all the codes attached into one pdf.
 - Students can use any coding language of their preference such as Matlab, Python, Julia, C++, etc. All the associated codes for the assignment problems should be attached at the end of each respective problem in your assignment submission. You can also use Jupyter notebook, Google colab, etc to do your assignment and submit it as one pdf.
 - You can use internet, any books, lecture notes, etc. Discussing with other students is perfectly fine and in fact is strongly encouraged. But plagiarism (copying solutions from others without putting your hard work) will never be tolerated.
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Problem 1: Fix the number of agents to be $N = 20$. Generate a connected graph using Erdos-Renyi model for N agents and some $p \in (0, 1)$. Assume random initial positions for all N agents in $\mathbb{R}^2$. Using formation control strategy, display your name in capital letters in $\mathbb{R}^2$, letter by letter in a sequence. For instance, my name is Venkat and this is how I would do:

- Starting from initial position vector $\mathbf{x}[0]$, agents should go to positions forming the letter **V**.
- Starting from positions corresponding to letter **V**, agents should go to positions corresponding to letter **E**
- Continue this letter sequence, until agents go to letter **T**.

Record your name formation sequence as an animation video through your simulation, host it in your github and simply give just the Github link containing both the code and the video as your solution. Ensure that your Github repository is made public before submitting it. For people having big lengthy names, choose the corresponding name which people use to call you on a regular basis - not the full entire name. For instance, if a student's full name is Manoj Kumar Sahu, and he goes commonly by simply Manoj, then he should simulate the sequence MANOJ.

Problem 2: Suppose that there is a target intruder which has to be tracked by a swarm of $N \in \mathbb{N}$ drones. Each of the N drones are equipped with a sensor and the drones are connected by a communication network $\mathcal{G}(\mathcal{V}, \mathcal{E})$. The initial location of the intruder is denoted by $z_0 \in \mathbb{R}^3$ is not known exactly and $z_0 \sim \mathcal{N}(\bar{z}_0, \Sigma_0)$, where mean $\bar{z}_0 \in \mathbb{R}^3$ and covariance $\Sigma_0 \in \mathbb{R}^{3 \times 3}$ and is positive semi-definite.



The intruder target is doing a random walk with dynamics given by

$$z(t+1) = z(t) + w, \quad \forall t \in \mathbb{N}, \quad (1)$$

where $w \sim \mathcal{N}(0, \Sigma_w)$ denotes the process noise (like the wind disturbance). Each sensor i is located at $x_i \in \mathbb{R}^3$ and assume that sensor i mounted on drone i knows its location x_i , and its measurement data $y_i(t)$ is given by the model $\forall t \in \mathbb{N}$ as

$$y_i(t) = x_i - z(t) + v_i. \quad (2)$$

Here v_i is a random sensor measurement noise with $v_i \sim \mathcal{N}(0, \Sigma_v^{(i)})$, where $\Sigma_v^{(i)} \succeq 0$ is also known to agent i . We assume that noises for different sensors (v_i, v_j with $i, j \in 1, \dots, N$ and $i \neq j$) are independent. Now let each drone with their respective sensor $i = 1, \dots, N$ take one measurement using (2) and obtain $y_i(t)$ for each time $t = [0, T_{\max}]$. Do the following:

1. Formulate the distributed state estimation problem of the intruder's location $z(t)$ as a consensus optimization problem, and specify each local objective function f_i .
2. Find the estimate of the intruder's state $\hat{z}(t)$ using distributed state estimation technique (distributed gradient descent or distributed gradient tracking) with the data collected by all the sensors $\{y_i(t), t \in [0, T_{\max}]\}_{i=1}^N$ via (2).
3. Assume and fix the number of drones $N \in \mathbb{N}$ (typically less than 20) and also generate a random graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ with N agents (using Erdos-Renyi model) containing a spanning tree (so that graph is connected) and plot it. Assume and fix the random quantities for $\bar{z}_0, \Sigma_0, \Sigma_w, \{x_i, \Sigma_v^{(i)}\}_{i=1}^N$ depicting above figure. Simulate and plot the fixed sensor (that is, drone) locations and actual true intruder location trajectory using (1) at every time for $t = [0, T_{\max}]$. Assume yourself some $T_{\max} \in \mathbb{N}$. Plot the computed estimate of the intruder's state trajectory $\hat{z}(t)$ for all $t = [0, T_{\max}]$ and check if your estimate matches well with the actual intruder state $z(t)$ and report the estimation error ($e(t) = \hat{z}(t) - z(t)$) too.

Problem 3: There are $N \in \mathbb{N}$ agents connected via an undirected connected graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$. They must collaboratively execute $M \in \mathbb{N}$ tasks where $M > N$. The set of all tasks is denoted by $\mathcal{T} := \{T_1, \dots, T_M\}$. Each agent $i \in \mathcal{V}$ is assigned a local cost C_{ij} for performing a task $T_j \in \mathcal{T}$ and thereby giving us a cost matrix $C \in \mathbb{R}^{N \times M}$. Further, each agent $i \in \mathcal{V}$ has a capacity constraint $b_i \in \mathbb{Z}_{\geq 0}$ which captures how many tasks that agent $i \in \mathcal{V}$ can take. Since, there is totally M tasks and each task should be assigned only once, we see that $\sum_{i=1}^N b_i = M$. Do the following:

1. Formulate the task allocation problem as an optimization problem with appropriate cost function and constraints. Your understanding of the problem and the way you formulate should be highlighted well. Give the ADMM updates for every agent $i \in \mathcal{V}$.

Hint: You should get a binary linear program (which is NP-hard to solve for large M, N). Hence, you will need to do convex relaxation which you can learn by yourself or come ask me.

2. Randomly choose the number of tasks M and number of agents N such that $M > N$. Fix them and generate an undirected connected graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ for the communication model (use Erdos-Renyi model). Randomly choose and fix the capacities b_i for every agent $i \in \mathcal{V}$ such that $\sum_{i=1}^N b_i = M$. Generate a random cost matrix $C \in \mathbb{R}^{N \times M}$ and ensure each element $C_{ij} > 0$. Plot the graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ and the result of task allocation using ADMM simulation. Fix the same (M, N) , play with different generated cost matrix $C \in \mathbb{R}^{N \times M}$ and plot the resulting task allocation and explain your observations.

Problem 4: Consider the LASSO problem

$$\min_x \left[\frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1 \right].$$

Write the ADMM iteration updates by solving and writing the closed form solution for x, z updates.